

# Timothy Chen

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## EDUCATION

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### Pepperdine University

*B.S. in Computer Science & Mathematics*

*Summa Cum Laude*

- Cumulative GPA: 3.992 / 4.000
- Coursework: Data Structures, Computer Systems, Machine Learning, Image Analysis, Computer Networking
- Awards: Multiple competitive departmental scholarships, over \$30k awarded in research grants (e.g. Northrop Grumman Endowed Scholarship, Keck Data Science Grant).

## EXPERIENCE

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### Amazon

June 2025 – Present

*Data Engineer - Security Issue Management*

*Seattle, WA*

- Designed a 100+ TB OLAP layer for AWS and SDO's unified internal security findings service using distributed processing (Spark), DynamoDB / Apache Iceberg for data management, and Athena for serverless analytics
- Owned and evolved core security issue data models that served 100+ internal security teams
- Built CI/CD for SageMaker ML workflows (CDK, Docker), standardizing training and inference environments
- Developed a ticket embedding and similarity retrieval pipeline with hybrid retrieval, ranking strategy, and feedback loop, improving mean triage time by 90% across 30+ weekly incidents

### Amazon

May 2024 – Aug. 2024

*Data Engineer Intern - Alexa*

*Seattle, WA*

- Provisioned cloud infra and deployed distributed processing pipelines with AWS Glue, TypeScript, AWS CDK
- Metrics invented and delivered to directors led to a 20% boost in Alexa's B2B conversions MoM

### Samsung

May 2023 – Aug. 2023

*Software Engineer Intern*

*Austin, TX*

- Redesigning Samsung Foundry's defect reporting tool for various in house equipment via Python on Streamlit
- Adopted by several core engineering teams, standardizing the defect reporting process

### Pepperdine Information Technology

Aug. 2022 – May 2024

*Software Engineer (Co-op)*

*Malibu, CA*

- Built full stack applications for University administration, including [esign.pepperdine.edu](https://esign.pepperdine.edu) and [sso.pepperdine.edu](https://sso.pepperdine.edu)
- Featured in Pepperdine IT's 2024 Annual Review [\[Article\]](#)

## PROJECTS

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### Cell Detector | *Python, Roboflow, scikit-learn, Google Colab*

- YOLO-based CNN detecting 1k+ plant cells per image with .94 F1; automated annotation for biology faculty
- This project was later published at ISVC '23 and presented at the 2023 ESA conference

### Intellectual Disability Balance Reference | *Applied Machine Learning*

- Built centile curves to track balance for people with intellectual disabilities, which were adopted in clinical practice
- Oral at the 2023 ISAPA conference in New Zealand was featured in Pepperdine's Newsletter [\[Article\]](#)

## SKILLS

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**Languages:** Python, TypeScript, C#, SQL, C++, Java, R, Assembly, JavaScript

**Tools & Frameworks:** IaC, CDK, AWS, .NET, MVC, ECS, PyTorch, Roboflow, Redshift, Postgres

**Hackathons:** Morgan Stanley CTG Hackathon Spring 2023, Alteryx Datathon 2023 Finalist

**Leadership:** TA - Formal Methods, Discrete Structures (5 semesters); Resident Life Advisor (4 semesters)